

EXPERIMENT-26 Fruit and Its Color Using Backtracking

Aim

To write a Prolog program to identify fruits and their colors using backtracking.

Objectives

- To understand facts in Prolog.
- To understand automatic backtracking.
- To retrieve multiple solutions for a query.

Algorithm

1. Define fruits and their colors.
2. Query a fruit to find its color.
3. Use ; to request additional solutions.
4. Prolog backtracks and returns all matching facts.

Flowchart

START

|

v

Define fruit facts

|

v

Query fruit/color

|

v

Search matching fact

|

v

More solutions?

/ \

Yes No

| |

v v

Backtrack STOP

Program

% Fruit and Color using Backtracking

fruit_color(apple, red).

fruit_color(apple, green).

fruit_color(banana, yellow).

fruit_color(orange, orange).

fruit_color(grapes, green).

fruit_color(grapes, purple).

fruit_color(mango, yellow).

fruit_color(mango, green).

fruit_color(guava, green).

fruit_color(watermelon, green).

Queries

?- fruit_color(apple, Color).

Output:

Color = red ;

Color = green ;

false.

?- fruit_color(mango, Color).

Output:

<pre>HelloWorld.pl × % Fruit and Color using Backtracking fruit_color(apple, red). fruit_color(apple, green). fruit_color(banana, yellow). fruit_color(orange, orange). fruit_color(grapes, green). fruit_color(grapes, purple). fruit_color(mango, yellow). fruit_color(mango, green). fruit_color(guava, green). fruit_color(watermelon, green). % Main program main :- write('Fruit and Color combinations:'), nl, fruit_color(Fruit, Color), write(Fruit), write(' -> '), write(Color), nl, fail. main. :- initialization(main).</pre>	I/O AI Agent STDIN Input for the program (Optional) Output: Fruit and Color combinations: apple -> red apple -> green banana -> yellow orange -> orange grapes -> green grapes -> purple mango -> yellow mango -> green guava -> green watermelon -> green
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Result

The program successfully retrieves fruit colors using Prolog backtracking.

Conclusion

Backtracking allows Prolog to automatically find multiple solutions to a query.